

GRAPH 6 – PEARSON CORRELATION MATRIX

The correlation matrix shows that Attendance ($r = 0.581$) and Hours_Studied ($r = 0.445$) are the two numeric attributes with the highest positive correlation with Exam_Score. Sleep_Hours and Physical_Activity exhibited near zero correlation, and were ranked as 0 by WEKA's InfoGain.

	<i>Hours_Studied</i>	<i>Exam_Score</i>	<i>Sleep_Hours</i>	<i>Previous_Scores</i>	<i>Tutoring_Sessions</i>	<i>Physical_Activity</i>	<i>Attendance</i>
Hours_Studied	1						
Exam_Score	0.445454954	1					
Sleep_Hours	0.010976689	-0.017021629	1				
Previous_Scores	0.024845782	0.175079087	-0.021750343	1			
Tutoring_Sessions	-0.014282264	0.156525185	-0.012216112	-0.013122326	1		
Physical_Activity	0.00462439	0.027824436	-0.000378065	-0.011273734	0.017732945	1	
Attendance	-0.009907859	0.581071863	-0.015917826	-0.020186103	0.014323509	-0.022434703	1

Figure 1 : Attendance (0.581) and Hours Studied (0.445) show the strongest positive correlations with Exam Score, while other variables such as Sleep Hours, Tutoring Sessions, and Physical Activity have negligible correlations, indicating minimal direct influence on exam performance.